



KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2009/2010

INSTITUTIONAL BASED PROGRAMME (APRIL SESSION)

EXAMINATION FOR THE DEGREE OF MASTER OF EDUCATION

ECT 530: ASSESSMENT AND LABORATORY MANAGEMENT IN CHEMISTRY

DATE: Friday 3rd September 2010 TIME: 9.00a.m – 12.00p.m

INSTRUCTIONS

Answer any three questions.

- Q1. A chemistry laboratory design is supposed to emphasize safety and flexibility. Discuss how this is realized in the following situations.
- a) Ensuring adequate ventilations.
 - b) Performing Teacher Demonstrations experiments.
 - c) Designing and performing students experiments.
 - d) Organizing for a theory lesson. [23marks]
- Q2. Safety is important when using chemistry laboratory. Discuss how the knowledge and application of the following can alleviate risks in chemistry laboratory.
- a) The Fire Triangle
 - b) Flash Points of chemicals
 - c) Threshold limits Values (TLVs) of chemicals
 - d) Has installations [23marks]

Q3. Designing practical experiments requires a lot of safety considerations. Using examples at school level, discuss the safety measures you would emphasize when designing and performing experiments that are likely to pose the following dangers.

- a) Sucking back
- b) Fire due to a flammable liquid
- c) Shattering of a glass container with reagents
- d) Emit toxic and corrosive fumes [23marks]

Q4. Assessment of chemistry should be linked to the main objectives of teaching chemistry. Discuss how the objectives guide assessment in the following cases.

- a) Chemistry assessment uses several theory papers
- b) Chemistry assessment used a practical paper at KCSE level
- c) Motivation is important in learning chemistry but is not directly assessed at school level.
- d) Teacher Tests have a role in the learning of chemistry although they are not recognized in the final grade. [23marks]

Q5. Practical work in chemistry is assessed using a practical paper.

- a) Outline the main types of skills emphasized at school level.
- b) Discuss the mode of assessment at KCSE level.
- c) Explain why there are three questions at KCSE level
- d) What improvement would you recommend to the process used by KNEC? [23marks]